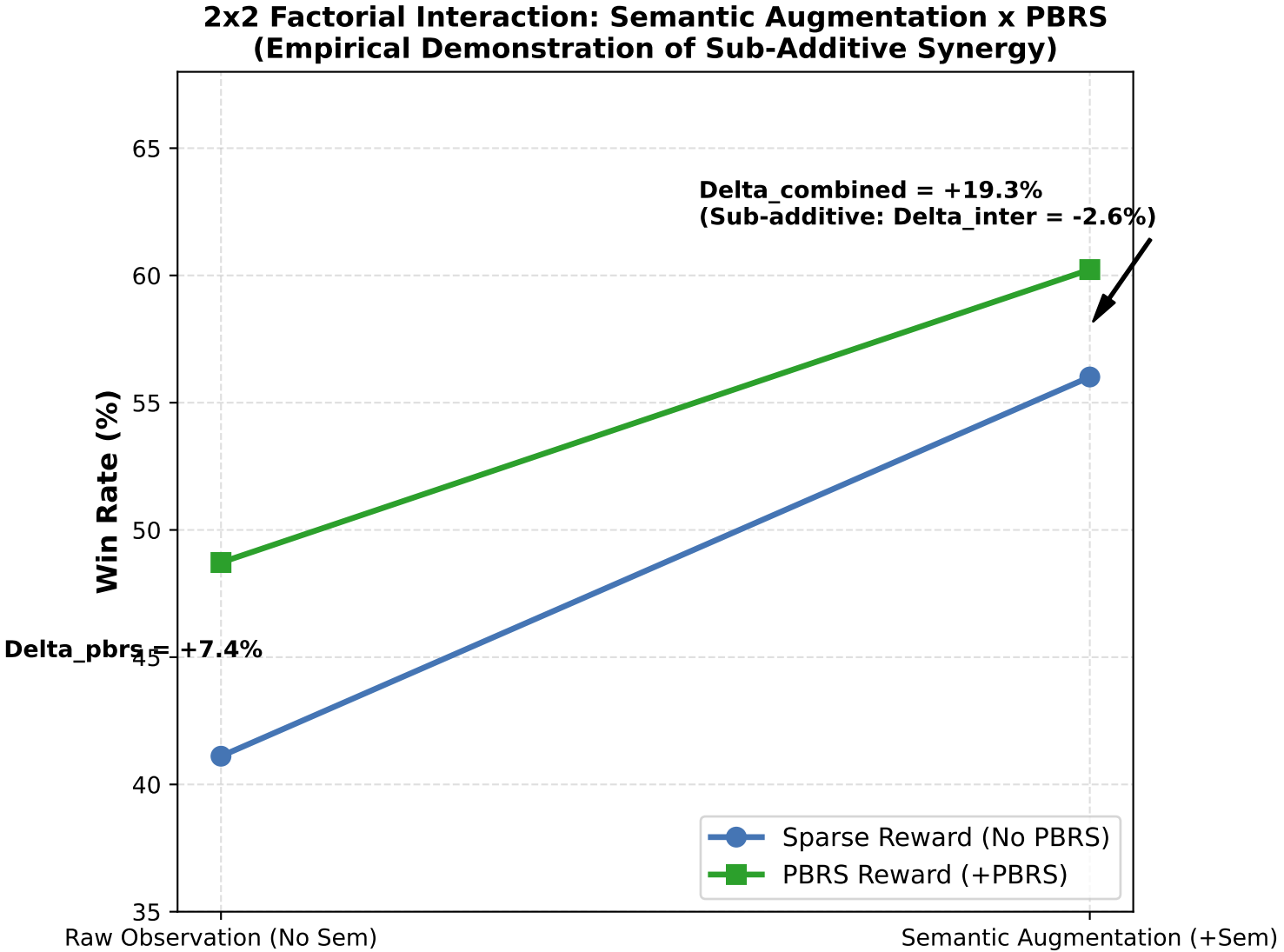
2x2 Factorial Interaction: Semantic Augmentation x PBRS
(Empirical Demonstration of Sub-Additive Synergy)
Win Rate (%)
65
60
55
50
45
40
35
Delta_combined = +19.3%
(Sub-additive: Delta_inter = -2.6%)
Delta_pbrs = +7.4%
Sparse Reward (No PBRS)
PBRS Reward (+PBRS)
Raw Observation (No Sem)
Semantic Augmentation (+Sem)